Example 4: Setup (Case II)

- Solve $(x^2 - x)y'' + (3x - 1)y' + y = 0$ near $x = 0$.
- Rewrite as $x(x - 1)y'' + (3x - 1)y' + y = 0$.
- Standard form: $P(x) = \dfrac{3x - 1}{x(x - 1)}$, $Q(x) = \dfrac{1}{x(x - 1)}$.
- Compute $b(x) = xP(x) = \dfrac{3x - 1}{x - 1}$; at $x = 0$, $p_0 = \dfrac{-1}{-1} = 1$.
- Compute $c(x) = x^2 Q(x) = \dfrac{x}{x - 1}$; at $x = 0$, $q_0 = 0$.
- Both $b(x), c(x)$ are analytic at $x = 0$, so $x = 0$ is a regular singular point.
- Indicial equation:
  $r(r - 1) + p_0 r + q_0 = 0 \implies r(r - 1) + r = 0 \implies r^2 = 0.$
- Roots: $r_1 = r_2 = 0$ (double root), so **Case II** applies.



Example 4: First Solution

- Substitute $y = \sum a_m x^{m+r}$ into $x(x-1)y'' + (3x-1)y' + y = 0$ directly.
- After simplification, the coefficient of x^{m+r-1} leads to

$$(m+r+1)^2 a_m = (m+r+1)^2 a_{m+1}.$$

- For $r = 0$: this reduces beautifully to $a_{m+1} = a_m$ for every $m \geq 0$.
- So all coefficients are equal: $a_m = a_0$ for every m.
- First solution: $y_1(x) = a_0 \sum_{m=0}^{\infty} x^m = \dfrac{a_0}{1-x}$.
- Taking $a_0 = 1$: $\quad y_1(x) = \dfrac{1}{1-x}$.
- We now find the second solution using reduction of order.



Example 4: Second Solution via Reduction of Order

- Since $y_1 = \frac{1}{1-x}$ is known exactly, use reduction of order:
  $y_2 = y_1 \int \frac{e^{-\int P\,dx}}{y_1^2}\,dx.$
- Partial fractions: $P(x) = \dfrac{3x-1}{x(x-1)} = \dfrac{1}{x} + \dfrac{2}{x-1}.$
- So $\int P\,dx = \ln|x| + 2\ln|x-1| = \ln\left[x(x-1)^2\right].$
- Hence $e^{-\int P\,dx} = \dfrac{1}{x(x-1)^2}.$
- Also $y_1^2 = \dfrac{1}{(1-x)^2} = \dfrac{1}{(x-1)^2}$, so $1/y_1^2 = (x-1)^2.$
- The integrand simplifies to $(x-1)^2 \cdot \dfrac{1}{x(x-1)^2} = \dfrac{1}{x}$
- So $\displaystyle\int \frac{1}{x}\,dx = \ln x$, giving $y_2(x) = y_1(x)\ln x = \frac{\ln x}{1-x}.$
- General solution: $y(x) = \dfrac{C_1 + C_2 \ln x}{1-x}.$
- This matches Case II form $y_2 = y_1 \ln x + (\text{series})$, Series=0.



Problem 5: Setup (Case III – No Logarithm)

- Solve near $x = 0$: $x^2y'' + xy' + \left(x^2 - \frac{1}{4}\right)y = 0$ (Bessel order $\frac{1}{2}$).
- Here $xP(x) = 1$ and $x^2Q(x) = x^2 - \frac{1}{4}$, both analytic at $x = 0$.
- Hence $x = 0$ is a regular singular point.
- Assume $y = \sum_{n=0}^{\infty} a_n x^{n+r}$ and substitute into the ODE.
- Indicial equation (lowest power x^r):

$$r(r-1) + r - \frac{1}{4} = 0 \implies r^2 = \frac{1}{4}.$$

- Roots: $r_1 = \frac{1}{2}$, $r_2 = -\frac{1}{2}$.
- Difference $r_1 - r_2 = 1$, a positive integer, so this is **Case III**.
- We must check whether the smaller root produces a logarithm or not
- Recurrence relation from matching x^{n+r}:

$$\left[(n+r)^2 - \frac{1}{4}\right]a_n + a_{n-2} = 0.$$

- We now test $r = r_2 = -\frac{1}{2}$ for a possible breakdown at $n = 1$.



Problem 5: Testing for a Logarithm

- With $r = -\frac{1}{2}$: recurrence becomes $\left[(n-\frac{1}{2})^2 - \frac{1}{4}\right] a_n = -a_{n-2}$.
- Simplify coefficient: $(n-\frac{1}{2})^2 - \frac{1}{4} = n^2 - n = n(n-1)$.
- At $n = 1$: coefficient is $1(0) = 0$, and we need $0 \cdot a_1 = -a_{-1} = 0$.
- Since the right side is also 0, the equation is satisfied for *any* value of a_1.
- Thus, **no logarithmic term is required**.
- a_1 becomes a second free constant, effectively generating the second series directly.
- For $n \geq 2$: $a_n = \dfrac{-a_{n-2}}{n(n-1)}$, same recurrence pattern as before.
- Choosing $a_1 = 0$ gives one purely even series; choosing $a_0 = 0, a_1 = 1$ gives an odd series.
- Both series combine into the single family $x^{-1/2} \sum a_n x^n$ with two free constants.
- This illustrates the "lucky" sub-case of Case III where the log term drops out.



Problem 5: Final Solution

- Working out the coefficients explicitly, the two independent solutions are

$$y_1(x) = x^{-1/2}\left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots\right) \cdot x = x^{1/2}\cos^{*}\text{-type series.}$$

- In closed form these series sum to elementary trigonometric functions:

$$y_1(x) = \sqrt{\frac{2}{\pi x}}\sin x, \qquad y_2(x) = \sqrt{\frac{2}{\pi x}}\cos x.$$

- These are the Bessel functions $J_{1/2}(x)$ and $J_{-1/2}(x)$.
- Both are perfectly well behaved except for the factor $x^{-1/2}$ at $x = 0$.
- Neither solution contains a logarithm, confirming our indicial test.
- General solution: $y = A\sqrt{\frac{2}{\pi x}}\sin x + B\sqrt{\frac{2}{\pi x}}\cos x$.
- This problem shows half-integer order Bessel equations always avoid logarithms.



Problem 6: Setup (Case III – With Logarithm)

- Solve near $x = 0$: $x^2y'' + xy' + (x^2 - 1)y = 0$ (Bessel's equation, order 1).
- Here $xP(x) = 1$ and $x^2Q(x) = x^2 - 1$, both analytic at $x = 0$.
- Hence $x = 0$ is a regular singular point.
- Assume $y = \sum_{n=0}^{\infty} a_n x^{n+r}$ and substitute into the ODE.
- Indicial equation: $r(r-1) + r - 1 = 0 \implies r^2 = 1$.
- Roots: $r_1 = 1,\ r_2 = -1$.
- Difference $r_1 - r_2 = 2 \in \mathbb{N}$, so this is **Case III**.
- Recurrence from matching x^{n+r}:

$$\left[(n+r)^2 - 1\right]a_n + a_{n-2} = 0.$$

- With $r = -1$: coefficient becomes $[(n-1)^2 - 1]a_n = n(n-2)a_n$.
- At $n = 2$: coefficient is $2(0) = 0$, but the right side requires $-a_0 \neq 0$.
- This is a genuine contradiction, so a **logarithmic term is forced** at $n = N = 2$.



Problem 6: First Solution ($r_1 = 1$)

- Using the larger root $r_1 = 1$, the recurrence $[(n+1)^2 - 1]a_n = -a_{n-2}$ has no breakdown.
- This gives $n(n+2)a_n = -a_{n-2}$ for $n \geq 2$, with $a_1 = 0$.
- All odd coefficients vanish; even coefficients follow $a_{2k} = \dfrac{(-1)^k a_0}{2^{2k} k!(k+1)!}$.
- First solution (with standard normalization $a_0 = \frac{1}{2}$):

$$y_1(x) = J_1(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!(k+1)!} \left(\frac{x}{2}\right)^{2k+1}$$

- This is the **Bessel function of the first kind**, order one.
- $J_1(x)$ is finite and well-defined at $x = 0$, in fact $J_1(0) = 0$.
- Since the smaller root failed (contradiction found on the previous slide) and the second solution must take the full logarithmic form.
- General form: $y_2(x) = CJ_1(x)\ln x + x^{-1}\sum_{n=0}^{\infty} b_n x^n$.
- The constant C and coefficients b_n are found by substitution.



Problem 6: Second Solution and Final Answer

- Substituting the trial form $y_2 = CJ_1(x)\ln x + x^{-1}\sum b_n x^n$ into the ODE and matching coefficients order by order determines C and all b_n uniquely.
- The standard result (after full computation) is the Bessel function of the second kind

$$Y_1(x) = \frac{2}{\pi}J_1(x)\ln x - \frac{1}{\pi}x^{-1} - \frac{1}{\pi}\sum_{k=0}^{\infty}(\cdots)x^{2k+1}.$$

- As expected, $C = \frac{2}{\pi} \neq 0$, confirming the logarithm is genuinely present.
- $Y_1(x) \to -\infty$ as $x \to 0^+$, unlike the finite $J_1(x)$.
- General solution: $y = A\,J_1(x) + B\,Y_1(x)$.
- Same integer root gap of pattern, but here the logarithm cannot be avoided.
- The deciding factor was whether the obstruction term vanished or not.



Problem 7: Setup and Solution (Case I Revisited)

- Solve near $x = 0$: $\quad 2x^2y'' - xy' + (1+x)y = 0.$
- Here $xP(x) = -\frac{1}{2}$ and $x^2Q(x) = \frac{1}{2}(1+x)$, both analytic at $x = 0$.
- So $x = 0$ is a regular singular point.
- Indicial equation: $2r(r-1) - r + 1 = 0 \implies 2r^2 - 3r + 1 = 0.$
- Factor: $(2r-1)(r-1) = 0 \implies r_1 = 1,\ r_2 = \frac{1}{2}.$
- Difference $r_1 - r_2 = \frac{1}{2}$, not a positive integer, so **Case I** applies.
- Recurrence: $[2(n+r)(n+r-1) - (n+r) + 1]a_n = -a_{n-1}.$
- For $r_1 = 1$: $a_n = \dfrac{-a_{n-1}}{2n(n+1) - n} = \dfrac{-a_{n-1}}{n(2n+1)}$, giving $y_1 = x\left(1 - \frac{x}{3} + \cdots\right).$
- For $r_2 = \frac{1}{2}$: $a_n = \dfrac{-a_{n-1}}{n(2n-1)}$, giving $y_2 = x^{1/2}\left(1 - x + \cdots\right).$
- General solution: $y = A\,y_1(x) + B\,y_2(x).$
- Both series converge for all $x > 0$ since $x = 0$ is the only singularity.



Problem 8: Setup and Solution (Case II Revisited)

- Solve the Cauchy-Euler equation: $x^2y'' + 3xy' + y = 0$.
- Here $xP(x) = 3$ and $x^2Q(x) = 1$, both trivially analytic (constants).
- So $x = 0$ is a (very simple) regular singular point.
- Indicial equation: $r(r-1) + 3r + 1 = 0 \implies r^2 + 2r + 1 = 0$.
- Factor: $(r+1)^2 = 0 \implies r_1 = r_2 = -1$ (repeated root).
- Since roots are equal, this is **Case II**: a logarithm is required.
- Because the equation is a pure Cauchy-Euler equation, the coefficient recurrence is trivial.
- All $a_n = 0$ for $n \geq 1$, so the first solution is simply $y_1(x) = x^{-1}$.
- The second solution follows the Case II template $y_2 = y_1 \ln x + \sum b_n x^{n-1}$.
- It reduces here to exactly $y_2(x) = x^{-1} \ln x$ (all $b_n = 0$ for $n \geq 1$).
- General solution: $y = Ax^{-1} + Bx^{-1} \ln x$.
- This example confirms the Case II log-term rule in closed form.

